

Abstract

This study updates the ILO's 2023 Global Index of Occupational Exposure to Generative AI (GenAI), incorporating recent advances in the technology and increasing user familiarity with GenAI tools. Using a representative sample from the 29,753 tasks in the Polish occupational classification system and a survey of 1,640 people employed in each 1-digit ISCO-08 groups, we collect 52,558 data points regarding perceive potential of automation for 2,861 tasks. We then compare this input with a survey and several rounds of Delphi-style discussions among a smaller group of international experts. Based on this process, we create a repository of knowledge about task automation that goes beyond national specificities and use it to develop an AI assistant able to predict scores for tasks in the technical documentation of ISCO-08.

Our 2025 scores are presented in a revised framework of four progressively increasing exposure gradients, with a new set of global estimates of employment shares exposed to GenAI. Clerical occupations continue to have the highest exposure levels. Additionally, some strongly digitized occupations have increased exposure, highlighting the expanding abilities of GenAI regarding specialized tasks in professional and technical roles.

Globally, one in four workers are in an occupation with some GenAI exposure. 3.3% of global employment falls into the highest exposure category, albeit with significant differences between female (4.7%) and male employment (2.4%). These differences increase with countries' income (9.6% female vs 3.5% male in Gradient 4 in HICs), and so does the overall exposure (11% of total employment in LICs vs 34% in HICs). As most occupations consist of tasks that require human input, transformation of jobs is the most likely impact of GenAI. Linking our refined index with national micro data enables precise projections of such transformations, offering a foundation for social dialogue and targeted policy responses to manage the transition.

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employed in each ISCO-08 1-digit group and from expert opinions. As the final step, we provide updated global, regional and income-based estimates of employment that has the potential to be transformed by Gen AI technology.

Our study specifically targets GenAI tools, isolating the effects of this recent family of AI from broader AI applications, such as the use of Machine Learning (ML) algorithms or image recognition in industrial production. There are several considerations that justify such selective focus. First, we respond to the growing need for a research tool that could provide a better picture of GenAI's potential impact on national labour markets, given the growing societal angst concerning this technology and the recent reports about increasing levels of adoption of GenAI at work (Bick et al., 2024; Faverio and Tyson, 2022; Grampp et al., 2023; Maison & Partners and ThinkTank, 2024; Pew Research Center, 2023; Rutgers, 2024). Second, while several indicators of occupational exposure to broader AI technologies are available to researchers (Nurski and Vansteenkiste, 2024), very few tools allow for a more detailed focus on GenAI (Nurski and Ruer, 2024). A new, more precise exposure index enables a closer alignment of the academic work on digital economy and labour with the significant interest in the GenAI technology in the media and in the public debate.¹ Third, the methodological blueprint provided by our research can be quite flexibly expanded to other types of AI in the future or focused more narrowly on specific subsets of digital technologies and sectors of their application.

This paper is the first in a series of upcoming publications based on ILO and NASK cooperation that strive to improve the methodology for assessing employment effects of AI, but also to document in detail the methods used, including the survey, expert verification process and the construction of the AI model used for predictions. Subsequent papers will present an assessment of the potential impact of GenAI on the Polish labour market, based on more detailed occupation and task-level data and using a mix of quantitative and qualitative methods. All publications will include the methodological blueprints and technical details of both the quantitative and qualitative components, including survey questionnaires and interview guidelines. By making these tools openly accessible, we hope to contribute to the development of an improved method for assessing potential impact of GenAI on jobs and to stimulate more detailed national-level studies, including in lower- and middle-income countries.

¹ For example, Felten et al. (2023) provided an updated version of their original exposure index from 2021, adjusting for the abilities of GenAI tools. As shown in their paper, the adjustment does not result in a major shift in occupations exposure and still makes it impossible to assess the abilities of this new technology in a selective manner. See Gmyrek et al. (2024) for a more detailed comparison of these scores.